

7	(a)	A photon is a discrete packet of electromagnetic energy. [1]			[1]	
	(b)	(i)	1.	$E_i = E_s + \frac{1}{2}mv^2$	[1]	
		(i)	2.	$p_i = p_s \cos \theta + mv \cos \alpha$	[1]	
		(ii)	Energy of photon, $E = hf$. As the scattered photon energy E_s is less than the incident photon energy E_i , the frequency of the scattered photon is smaller than that of incident photon. [1] As the wavelength is inversely proportional to its frequency, the scattered photon will have a wavelength greater than that of the incident photon. [1]			[2]
	(c)		Actual value, $\cos 75^\circ = 0.2588$ Check that $\cos 80^\circ = 0.1736$, $\cos 70^\circ = 0.3420$ Using $\Delta R = (R_{\max} - R_{\min}) / 2$ $\Delta(\cos \theta) = (\cos 70^\circ - \cos 80^\circ) / 2 = 0.08$ (1 s.f.) Therefore $\cos \theta = 0.26 \pm 0.08$			[2]
	(d)	(i)	Calculate $\cos \theta$ Correctly plot the two data points to within half smallest square			[1] [2]
		(ii)	Correct best fit line			[1]
		(iii)	$\Delta\lambda = k - k \cos \theta$ Method 1: gradient of graph gives negative k [1] Method 2: y-intercept of graph gives k [1]			[2]
		(iv)	Correct values for k based on graphs. [1] Correct unit for k . [1]			[2]

	(e)	(i)	Energy of Incident beam 19.0 keV $E = \frac{hc}{\lambda}$ Wavelength, $\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{(19.0 \times 10^3)(1.6 \times 10^{-19})} = 6.53 \times 10^{-11} \text{ m}$ Using $\lambda' = \lambda + \frac{h}{mc}(1 - \cos \theta)$, $\lambda' = 6.56 \times 10^{-11} \text{ m}$ Alternatively, accept $\Delta\lambda = k - k \cos \theta$ and using the value of k obtained in (d)(iv).	[1] [1]
		(ii)	As the binding energy of an electron of a carbon atom is negligible compared with the energy of the incident photons (19 keV). Therefore it is justified to assume that the electrons are free.	[1] [1]